Elevating All Zero-Shot Sketch-Based Image Retrieval Through Multimodal Prompt Learning

Mainak Singha[1], Ankit Jha[1,2], Divyam Gupta[1], Pranav Singla[1], Biplab Banerjee[1]
[1]Indian Institute of Technology Bombay, India, [2]INRIA, Grenoble, France

EUROPEAN CONFERENCE ON COMPUTER VISION
MILANO 2024

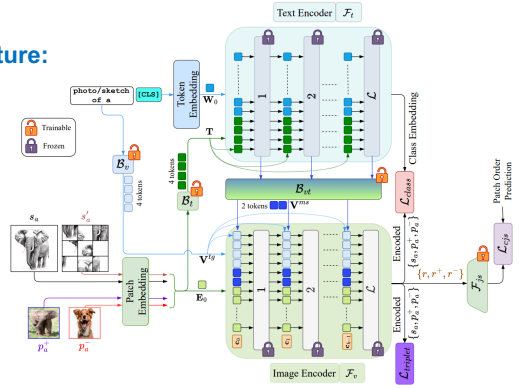
Motivation:

- Existing methods often rely on simplistic, one-dimensional prompting approaches, failing to fully leverage CLIP's visual-textual fusion. This leads to suboptimal performance in ZS-SBIR task due to missed opportunities for rich, complementary insights from CLIP's dual pathways.
- While multi-modal methods improve semantic alignment by prompting both visual and textual pathways simultaneously, they often struggle with integrating prompts effectively and maintaining textual adaptability. This results in reduced semantic depth and less sensitivity to visual nuances.
- The patch shuffling strategy, which improves shape equivalence in images and sketches, may lead to overfitting due to indiscriminate matching. Hence, it needs attention to match patch-permuted sketches with semantically corresponding, unshuffled photos to better align embeddings with the actual structure and objects in the images.

Our Proposal:

- We introduce SpLIP, a new deep multi-modal prompt tuning model designed for zero-shot and few-shot sketch-based image retrieval tasks, enhancing CLIP's text and vision encoders through systematic cross-modal mutual guidance.
- We advance the traditional cross-modal triplet loss function in ZS-SBIR by integrating an adaptive margin scheme and introduce a conditional cross-modal jigsaw task to better match sketch and photo pairs.
- Extensive experiments on Sketchy-Ext, TU-Berlin-Ext, and QuickDraw-Ext datasets show that SpLIP outperforms existing methods, achieving significant improvements across all metrics.

Architecture:



Proposed Methodology:

- **Vision-guided Deep Textual Prompting:**

$$[_, W_l] = F_t^l([T_{l-1}; W_{l-1}]) \in \mathbb{R}^{M \times d_t}, \qquad l = 1,2,\ldots,L$$

where, $T_l = T = B_t(E_0)$; B_t = visual-to-textual mapping block, W = word embeddings, E_0 = visual patch embeddings.

- **Text-guided Deep Visual Prompting:**

$$[c_l, _; E_l] = F_v^l([c_{l-1}, V_{l-1}; E_{l-1}]) \in \mathbb{R}^{1+N \times d_v}; \qquad V_{l-1} = [V_{l-1}^{tg}, V_{l-1}^{ms}]$$

where,

$$V_{l-1}^{ms} = \{V_{k_{l-1}}^{ms} \in \mathbb{R}^{d_v}\}_{k=1}^{n} = B_{vt}\left(F_t^l([T_{l-1}; W_{l-1}])\right) \in \mathbb{R}^{n \times d_v}$$

and $V_l^{tg} = V^{tg} = B_v(W')$; B_v = textual-to-visual mapping block

B_{vt} = vision-text conjunction block

- **Conditional Cross-modal Jigsaw Solver:**

✓ To resolve the jigsaw puzzle, we introduce a transformer-based jigsaw-solver network by leveraging the information of extracted feature embeddings.

✓ We propose a conditional cross-modal jigsaw loss to solve the fine-grained feature association problem.

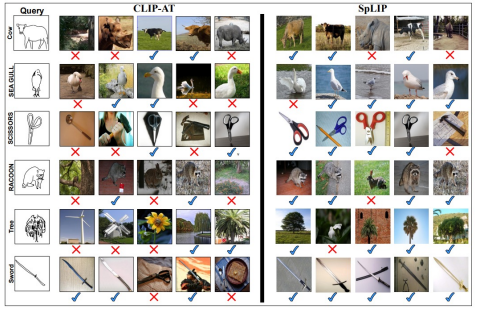
Experiments and Results:

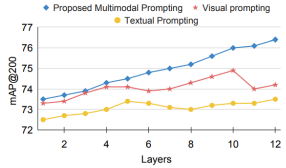
Table 1: Comparison for categorical ZS-SBIR.

	Methods		Sketchy-1-Ext [11] mAP@all	Sketchy-1-Ext [11] P@100	Sketchy-2-Ext [57] mAP@200	Sketchy-2-Ext [57] P@200	TU-Berlin-Ext [27] mAP@all	TU-Berlin-Ext [27] P@100	QuickDraw-Ext [7] mAP@all	QuickDraw-Ext [7] P@200
CNN	CVAE [57]	ECCV'18	19.6	28.4	22.5	33.3	00.5	00.1	00.3	00.3
CNN	CC-DG [31]	CVPR'19	31.1	46.8	-	-	24.7	39.2	-	-
CNN	Doodle [7]	CVPR'19	-	-	46.1	37.0	11.0	12.1	07.5	06.8
CNN	SEM-PCYC [11]	CVPR'19	34.9	46.3	45.9	37.0	29.7	42.6	17.7	18.4
CNN	SAKE [28]	ICCV'19	54.7	69.2	49.7	59.8	47.5	59.9	-	-
CNN	Styleguide [12]	TM'20	37.6	48.4	35.8	40.0	25.4	35.5	-	-
CNN	OCEAN [66]	ICME'20	-	-	-	-	33.3	46.7	-	-
CNN	DSN [56]	IJCAI'21	58.3	70.4	-	-	48.4	59.1	-	-
CNN	TCN [54]	TPAMI'21	61.6	76.3	51.6	60.8	49.5	61.6	14.0	29.8
CNN	BDA [3]	NC'22	43.7	51.4	55.6	45.8	37.4	50.4	15.4	35.5
CNN	Sketch3T [41]	CVPR'22	-	-	57.9	64.8	50.7	67.1	-	-
ViT	TVT [52]	AAAI'22	64.8	79.6	53.1	61.8	48.4	66.2	14.9	29.3
ViT	PSKD [55]	ACM MM'22	68.8	78.6	56.0	64.5	50.2	66.2	15.0	29.8
ViT	SaA [36]	-	67.1	76.2	53.5	63.0	49.5	60.8	14.8	-
ViT	ZSE-RN [26]	CVPR'23	69.8	79.7	52.5	62.4	54.2	65.7	14.5	21.6
ViT	ZSE-Ret [26]	CVPR'23	73.6	80.8	50.4	60.2	56.9	63.7	14.2	20.2
CLIP	CLIP-AT [39]	CVPR'23	-	-	72.3	72.5	65.1	73.2	20.2	38.8
CLIP	TLT [59]	MMTA'23	77.9	84.3	66.1	73.0	61.5	69.5	27.8	-
CLIP	Sherry [8]	-	74.1	83.5	61.6	69.5	54.1	66.4	18.0	29.8
CLIP	MARL [29]	WACV'24	-	-	69.1	75.5	70.5	77.7	32.7	42.5
	SpLIP		**80.2**	**86.7**	**76.4**	**77.3**	**73.1**	**78.2**	**34.2**	**44.6**

	Methods		Sketchy-Ext mAP@200	Sketchy-Ext P@200	TU-Berlin-Ext mAP@all	TU-Berlin-Ext P@100
CNN	SEM-PCYC [11]	CVPR'19	-	-	19.2	29.8
CNN	OCEAN [67]	ICME'20	-	-	31.2	34.1
CNN	BDA [3]	NC'22	22.6	33.7	25.1	35.7
ViT	SaA [36]	-	-	-	29.0	38.1
ViT	ZSE-Ret [26]	CVPR'23	-	-	46.4	48.5
ViT	ZSE-RN [26]	CVPR'23	-	-	43.2	46.0
ViT	STL [15]	AAAI'23	63.4	53.8	40.2	49.8
CLIP	CLIP-AT [39]	CVPR'23	55.6	62.7	60.9	63.8
CLIP	MARL [29]	WACV'24	62.3	68.5	62.6	67.8
	SpLIP		**68.2**	**74.5**	**66.7**	**70.3**

Methods	Sketchy Acc@1	Sketchy Acc@5
CrossGrad [45]	13.40	34.90
CC-DG [31]	22.60	49.00
SketchPVT [40]	30.24	51.65
CLIP-AT [39]	28.68	62.34
MARL [29]	29.96	58.53
SpLIP	**33.45**	**66.71**





Summary:

- We present SpLIP, that features via multi-modal prompt learning for improved cross-modal knowledge sharing, dynamic margin adjustment using CLIP's textual embeddings, and a conditional cross-modal jigsaw task for better sketch-photo patch-level alignment.
- Superior performance of SpLIP across various benchmarks, that could be extended to broader vision-language domains to enhance the understanding of visual semantics in future works.

